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ORIGINAL ARTICLE

Predicting Post-ERCP Pancreatitis Using Machine Learning: Risk Stratification and Feature Importance Analysis

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ABSTRACT

Background: Despite advances in understanding the risk factors of post-endoscopic retrograde cholangiopancreatography (ERCP) pancreatitis (PEP), this adverse event remains frequently unpredictable. This study aims to develop a machine learning (ML) model to predict PEP risk.

Methods: Data were collected from a prospective ERCP registry on patients with naïve papilla who underwent ERCP between 2022 and 2024. CatBoost and eXtreme Gradient Boosting algorithms were trained to estimate PEP risk and the performance of the resulting models was assessed using the area under the receiver operating characteristic (AUC) with 10-fold cross-validation.

Results: Of 1330 screened patients, 1190 met the inclusion criteria, and 170 (14.3%) developed PEP. The best-performing algorithm was CatBoost, which consisted of eight features: age, sex, normal papilla morphology, pancreatic duct cannulation, difficult cannulation, abnormal bilirubin levels, common bile duct diameter, and successful stone extraction. This model achieved an AUC of 68.8% (70.4% sensitivity, 67.2% specificity, 26.5% positive predictive value, and 92.0% negative predictive value). The CatBoost model effectively stratified patients into low-, intermediate-, and high-risk groups, with corresponding PEP incidences of 5.7%, 21.0%, and 40.0%, respectively.

Conclusions: ML is highly promising for prediction of PEP. Future studies should focus on multicenter data, inclusion of multimodal data, severity risk-stratification, and real-time application.

1 | Introduction

Endoscopic retrograde cholangiopancreatography (ERCP) is a key diagnostic and therapeutic procedure for managing pancreatobiliary disorders. Despite its utility, the procedure carries potential adverse events, even in experienced hands [1, 2]. The most common adverse event associated with ERCP is post-ERCP pancreatitis (PEP), which occurs in approximately 10.2% of patients, with an increased incidence of 14.1% among high-risk individuals, with a paramount associated morbidity and cost [3]. Several predictive scoring tools to stratify patients based on their risk of developing PEP have been developed [4–6], but these are rarely implemented in clinical practice, and PEP remains frequently unpredictable. One of the major limitations of most of these predictive tools to date has been the use of traditional statistical methodologies, often using multivariate logistic

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regression analysis, which cannot handle interdependent and nonlinear relationships between multiple PEP predictive factors. Meanwhile, having multiple PEP risk factors simultaneously can amplify both the likelihood and severity of PEP, producing a combined effect that exceeds the impact suggested by each individual risk factor alone [7]. Furthermore, many of these existing risk scores were derived and validated in specific and homogenous clinical settings. This has raised concerns about their generalizability and performance across diverse patient populations and practice environments, indicating a need for more robust external validation. Therefore, recent guidelines on the prevention of PEP highlight the need for novel individualized PEP risk assessment using both patient-related and procedure-related features [8].

In recent years, machine learning (ML) has achieved amazing advances in a range of sectors, including healthcare settings [9]. ML is a broad field of artificial intelligence that combines statistics and computer science. Repeated model iterations conducted by a machine can lead to gradually improved performance in carrying out a given task. ML can examine large amounts of data, assess hidden complicated nonlinear connections between variables, and choose the most important characteristics to build a model [10]. ML has developed successful evaluation and prediction skills in a variety of tough clinical activities by creating appropriate models with enhanced characteristics [11, 12]. In this study, we aimed to build and evaluate the performances of ML models to predict PEP probability via two tree-based classifiers, eXtreme Gradient Boosting (XGB) and CatBoost (CB). In addition, the identification of relevant procedure-related and patient-related variables to predict PEP risk was conducted. This would provide a more individualized approach to patient management and procedural planning, ultimately minimizing the incidence of PEP and improving patient outcomes.

2 | Patients and Methods

2.1 | Study Design

This study is a retrospective analysis of a prospective ERCP registry (IR.SBMU.RIGLD.REC.1398.043), on consecutive patients with naïve major duodenal papilla (MDP) who were referred for ERCP at the endoscopy unit of Taleghani Hospital, a tertiary referral center for pancreatobiliary disorders, Tehran, Iran. All patients with naïve MDP who underwent ERCP during the period from March 2022 to October 2024 were considered to be included in this study. The exclusion criteria were as follows: (1) age under 18 years, (2) pregnancy or breastfeeding, (3) surgically altered upper gastrointestinal anatomy, (4) unsuccessful biliary cannulation, (5) chronic pancreatitis, (6) ampullary cancer, and (7) pancreatic duct (PD) stent placement.

2.2 | ERCP Procedures

All procedures were performed by three experienced endoscopists with at least 10 years of ERCP experience, who perform more than 500 therapeutic ERCPs annually and achieve a > 90% success rate in selective deep biliary cannulation using standard techniques. A side-viewing therapeutic duodenoscope (JF-240 or JF-260V; Olympus, Tokyo, Japan) was utilized for the procedures. No trainees were involved in the procedures.

All patients received 100 mg of rectal indomethacin or diclofenac within a half-hour before the procedure and were hydrated with either normal saline or lactated ringer solution following the standard protocol (1.5 cc/kg/h during and for 8 h post-procedure). Patients were monitored for at least 6 h postanesthesia and hospitalized for at least 24 h. Serum amylase and lipase were measured at 6 and 24 h after ERCP. A follow-up phone call was made 30-days post-discharge to investigate any delayed adverse events.

2.3 | Definitions and Classifications

PEP was defined in accordance with the revised Atlanta classification with at least two of the following items: (I) new-onset abdominal pain after the procedure compatible with acute pancreatitis, (II) elevation of pancreatic enzyme ≥ 3 times the upper limit of normal, and (III) evidence of acute pancreatitis on imaging [13]. Serum total bilirubin > 1.2 mg/dL is considered abnormal.

The MDP morphology was defined according to the Haraldsson classification: (I) regular or normal papilla, (II) small papilla, (III) creased or long papilla, and (IV) protruding or bulging papilla [14]. Difficult biliary cannulation was defined according to the European Society of Gastrointestinal Endoscopy (ESGE) statement: (I) > 5 min of cannulation attempts, (II) > 5 contacts with the papilla, or (III) ≥ 2 inadvertent PD cannulation/opacification [15]. Primary needle-knife fistulotomy (NKF) was defined as the NKF as the initial cannulation technique, while late NKF was defined as the NKF after 5 min of unsuccessful cannulation attempts using the standard transpapillary technique. The technical details of cannulation approaches are explained in our previous studies [16, 17].

2.4 | Software and Packages

The model development and evaluation were performed using Python 3.12.7 (Python Software Foundation) in Anaconda open-source environment with Sci-kit learn (V. 1.5.1), catboost (V. 1.2.3), xgboost (V. 2.1.1), and SHAP (V. 0.46.0) packages.

2.5 | Imputation and Feature Selection

Features with rare occurrences (< 1%) were excluded from the model. The only variable with missing values was body mass index (BMI) (50/1190, 4.2%), which missing values were imputed with the mean BMI of the training set (75% of the patients under a 10-fold cross-validation [CV]).

For feature selection, variables were first filtered by a univariate analysis and those with $p < 0.2$ were retained. Then, a spearman correlation coefficient was computed to assess the multicollinearity between the variables. For correlation coefficients ≥ 0.7 or ≤ −0.7, only the variable with better predictive performance was kept.

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To select the best features in order to mitigate overfitting, different sets of features were tested. These sets of features were selected with a combined approach as the suggestion of the experienced endoscopists and the results of the feature selection analysis by recursive feature elimination with cross-validation (RFECV) and the least absolute shrinkage and selection operator (LASSO).

2.6 | Predictive Models

To develop the predictive models, we used two tree-ensemble algorithms: CB and XGB. They were chosen because of their adaptability and resilience in controlling tabular datasets, as well as their ability to capture nonlinear connections between parameters [18]. The key difference lies in the handling of categorical features: XGB requires heavy preprocessing, such as label encoding, while CB natively handles categories and most of the time gives better results with less preprocessing. Also, CB has an ordered boosting technique that reduces overfitting; XGB uses a more traditional gradient boosting approach with regularization.

To perform the predictions, we first used all of the eligible features as the input, and then a certain combination of features to enhance the performance of the models. We performed a grid search to find the model parameter that achieved a higher area under the receiver operating characteristic (AUC) for the training set. The number of trees and the interaction depth of the tree were tuned under 10-fold CV over a grid of 100, 300, and 500 trees, a maximum depth of 3, 4, and 5, and the learning rate of 0.001, 0.01, 0.1, and 0.3.

At the end of the CV process, for each set of features, mean AUC and 95% confidence interval (CI) on the 10 CV splits were calculated for CB and XGB. Finally, the model was used on the test set. The diagnostic metric of models, including AUC, sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), accuracy, precision, and F1 score, were calculated for evaluating the model performance on the test set. The mean and 95% CIs of the metrics of test sets were calculated by bootstrap method with 50 iterations. Receiver operating characteristic (ROC) curves were also plotted. The predicted probability threshold was chosen to maximize Youden's index. Two risk cut-offs were defined based on risk probability distributions and expert knowledge, dividing patients into low-, intermediate-, and high-risk groups. The cut-off values were defined using the posterior probabilities from the best model with the explicit goal of creating clinically distinct strata with meaningfully different PEP rates.

2.7 | Model Explainability

In order to understand the complex interactions of features influencing the model's predictions, we employed the SHapley Additive exPlanations (SHAP) approach. This method is an attribution technique assigning importance values (SHAP values) to each feature, indicating their contribution to the final prediction. In a SHAP summary plot, the *x*-axis shows the SHAP value, which measures the impact of each feature on the output of the model, and the *y*-axis shows the list of features used in the model. Positive SHAP values mean the feature increases the likelihood of PEP, while negative values mean the feature decreases the likelihood of PEP.

2.8 | Comparing Performance of the Best ML and Experts' Opinion Models

Data were prospectively collected from ERCP procedures performed at Taleghani Hospital during January and February 2025. The inclusion and exclusion criteria were consistent with those applied in the main study. Each case was evaluated using two separate models: (I) the best-performing ML model developed in the main study and (II) a voting classifier composed of three expert endoscopists who performed the procedures. For the experts' opinion model, all available features from the ERCP registry (including those excluded during the development of the ML model) were provided to each endoscopist. Each expert independently assessed the input data and classified the likelihood of PEP occurrence (yes or no). The final decision of the voting classifier was based on majority agreement, requiring at least two out of three experts to concur. Predictive performances of the ML and experts' opinion models were compared using AUC.

3 | Results

3.1 | Study Characteristics

Records of 1330 patients were retrospectively reviewed and finally, 1190 patients met the inclusion criteria (Figure 1a). Of them, 170 (14.3%) developed PEP. Among these 170 patients, 24 (14.1%) had moderate to severe cases, which corresponds to 2.0% of the overall cohort. Table 1 shows the characteristics of included patients.

The feature selection process is presented in Figure 1b in detail. A total of 39 possible risk factors of PEP were identified through the electronic health records. Among them, sphincter of Oddi dysfunction was excluded from the model because of its rare occurrence (10 cases, 0.84%). The univariate analysis led to exclusion of the 17 other variables with $p > 0.2$ (Figure S1). Then, Pearson correlation coefficient between the remaining variables revealed a significant correlation between difficult cannulation and number of cannulation attempts ($r = 0.79$), and biliary stricture and choledocholithiasis ($r = -0.75$) (Figure S2). Finally, a combination of the remaining 19 variables was used to develop 25 unique predictive models.

3.2 | XGB Model

The same combinations of features used for the CB model were also used for the XGB model. In its best performance in CV, the XGB model achieved an AUC of 0.751 (95% CIs 0.688–0.813) in 10-fold CV (model 1). However, when it was applied to the test set, the AUC significantly decreased to 0.613 (95% CIs 0.566–0.666) (unstable model). The best performance on the test set was observed in model 10; with an AUC of 0.662 (95% CIs 0.639–0.682), sensitivity 0.692 (95% CIs 0.633–0.744), and

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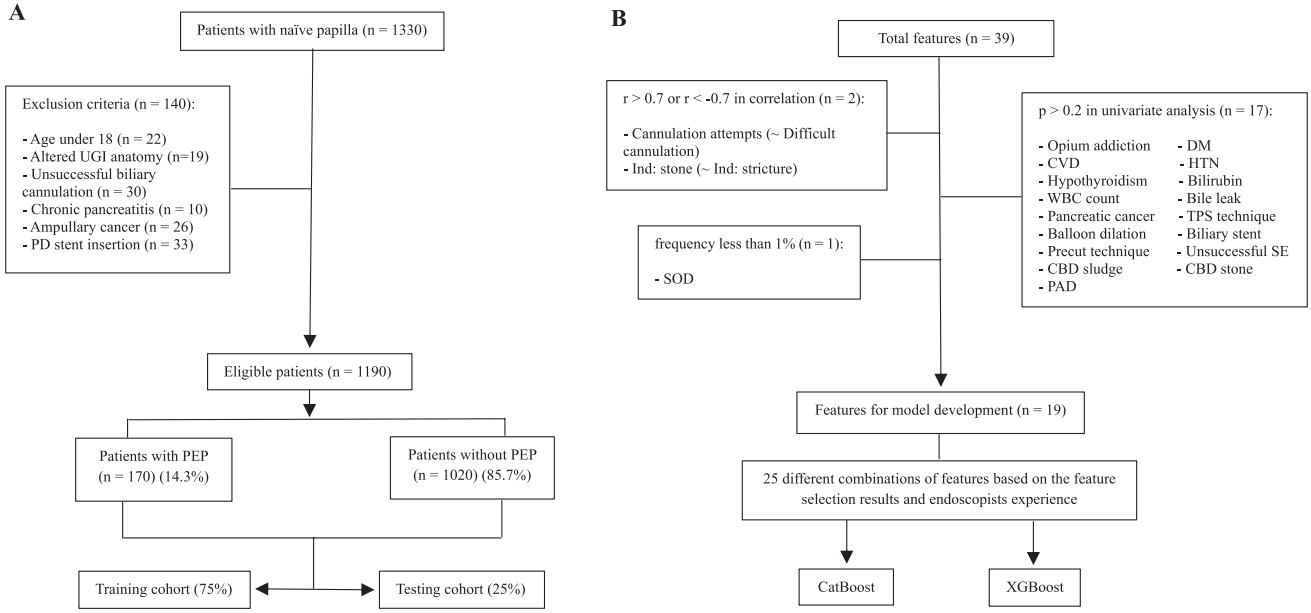


FIGURE 1 | Study algorithm. (A) Patients enrollment based on the inclusion and exclusion criteria; (B) selection of features process. CBD, common bile duct; CVD, cardiovascular disease; DM, diabetes mellitus; HTN, hypertension; Ind, indication; PAD, periampullary diverticulum; PD, pancreatic duct; PEP, post-ERCP pancreatitis; SE, stone extraction; TPS, trans-pancreatic sphincterotomy; UGI, upper gastrointestinal; WBC, white blood cell.

specificity 0.632 (95% CIs 0.585–0.678). Its performance in CV revealed an AUC of 0.721 (95% CIs 0.682–0.760) (stable model). PD cannulation, primary NKF technique, difficult cannulation, sex, and sphincterotomy, stone size, serum bilirubin levels, and biliary stricture were recognized as the most relevant predictive features (Figure S3). The results of the other 23 models with detailed metrics are presented in Table S1.

3.3 | CB Model

Twenty-five different combinations of features tested for CB model are shown in Table S2. In its best performance in CV, the CB model achieved an AUC of 0.775 (95% CIs 0.737–0.813) in 10-fold CV (model 2). However, when it was applied to the test set, the AUC significantly decreased to 0.665 (95% CIs 0.628–0.689) (unstable model). The best performance on the test set was observed in model 16; with the AUC of 0.688 (95% CIs 0.647–0.718), sensitivity 0.704 (95% CIs 0.643–0.738), specificity 0.672 (95% CIs 0.614–0.708), PPV 0.265 (95% CIs 0.236–0.292), and NPV 0.920 (95% CIs 0.889–0.948). Its performance in CV revealed an AUC of 0.710 (95% CIs 0.662–0.757) (stable model) (Figure 2a–c). Common bile duct (CBD) diameter, difficult cannulation, PD cannulation, age, serum bilirubin levels, sex, successful stone extraction, and normal MDP morphology were recognized as the most relevant predictive features (Figure S4).

As the CB16 model showed the best predictive performance, we used posterior probabilities of this model to classify the test set patients into the three groups of low-, intermediate-, and high-risk. The relevant cut-off values were <0.16, 0.16–0.45, and ≥ 0.45, respectively. In the test set, the incidence of PEP was 5.7% in the low-risk group (8 of 140 patients), 21.0% in the intermediate-risk group (30 of 143 patients), and 40.0% in the high-risk group (4 of 10 patients) (Figure 2d).

3.4 | Model Explainability

In order to interpret the effect of certain values of each feature on the output of the model, the SHAP method was employed. Figure 3 shows the list of all relevant features and their relative effect on PEP prediction for each patient. By examining the plot, we found some features to be associated with an increased risk of developing PEP and some protective features. Notably, features such as normal serum bilirubin level, PD cannulation, younger age, difficult cannulation, and female sex were linked to a higher risk of PEP, while features like successful stone extraction, larger stone size, primary NKF technique, and smoking were associated with a lower risk of PEP.

3.5 | Comparing Performance of the Best ML and Experts' Opinion Models

Data were collected from 102 ERCP procedures performed at the Taleghani Hospital including 19 cases of PEP (18.6%). Figure 4a shows the ROC curve for the PEP prediction of the best ML and experts' opinion models, with an AUC of 0.72 and 0.62, respectively. The best ML model had a sensitivity of 68.4%, specificity of 67.5%, PPV of 32.5%, and NPV of 90.3%. The experts' opinion model had a sensitivity of 42.1%, specificity of 81.9%, PPV of 34.8%, and NPV of 86.1% (Figure 4b).

4 | Discussion

PEP is a common adverse event following ERCP, constituting a significant source of morbidity and cost. There are multiple preventive strategies for PEP, including nonsteroidal anti-inflammatory drugs, aggressive fluid resuscitation, and PD stenting [19]. The ability to recognize high-risk patients for developing PEP prior to or early into the procedure may promote endoscopists to adopt

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TABLE 1 | Characteristics of enrolled patients.

Variables	Included patients ($n=1190$)	No PEP ($n=1020$)	PEP ($n=170$)	p
Age (years)	59.1 ± 17.7	59.6 ± 17.6	56.2 ± 17.8	0.021
Female sex	625 (52.5%)	517 (50.7%)	108 (63.5%)	0.002
Smoker	226 (19%)	204 (20.0%)	22 (12.9%)	0.030
Opium addiction	169 (14.2%)	146 (14.3%)	23 (13.5%)	0.786
Hypertension	280 (23.5%)	242 (23.7%)	38 (22.4%)	0.696
Diabetes mellitus	177 (14.9%)	156 (15.3%)	21 (12.4%)	0.318
Cardiovascular disease	105 (8.8%)	89 (8.7%)	16 (9.4%)	0.770
Hypothyroidism	27 (2.3%)	25 (2.5%)	2 (1.2%)	0.302
Body mass index	25.4 ± 4.7	25.5 ± 4.7	24.9 ± 4.7	0.129
Total bilirubin (mg/dL)	5.1 ± 6.4	5.1 ± 6.3	5.1 ± 6.9	0.244
Abnormal bilirubin	824 (69.2%)	726 (71.2%)	98 (57.7%)	< 0.001
White blood cell count ($10^3/\mu L$)	8.0 ± 3.4	8.0 ± 3.4	7.9 ± 3.3	0.735
Indication				
Choledocholithiasis	965 (81.1%)	836 (82.0%)	129 (75.9%)	0.061
Biliary stricture	147 (12.3%)	115 (11.3%)	32 (18.8%)	0.006
Pancreatic cancer	30 (2.5%)	27 (2.6%)	3 (1.8%)	0.790
Bile leak	19 (1.6%)	15 (1.5%)	4 (2.3%)	0.335
Others	29 (2.5%)	27 (2.6%)	2 (1.2%)	0.413
MDP morphology				
Normal	539 (45.3%)	447 (43.8%)	92 (54.1%)	0.013
Small	52 (4.4%)	41 (4.0%)	11 (6.5%)	0.148
Long	314 (26.4%)	278 (27.3%)	36 (21.2%)	0.096
Bulging	81 (6.8%)	76 (7.5%)	5 (2.9%)	0.031
Periampullary diverticulum	204 (17.1%)	178 (17.4%)	26 (15.3%)	0.477
Cannulation technique				
Standard transpapillary	512 (43.0%)	449 (44.0%)	63 (37.1%)	0.090
Primary NKF	300 (25.2%)	280 (27.5%)	20 (11.8%)	< 0.001
Late NKF	240 (20.2%)	182 (17.8%)	58 (34.1%)	< 0.001
TPS	80 (6.7%)	62 (6.1%)	18 (10.6%)	0.030
Precut papillotomy	36 (3.0%)	31 (3.0%)	5 (2.9%)	0.945
Others	22 (1.9%)	16 (1.6%)	6 (3.5%)	0.076
Cannulation attempts				
Up to 5	652 (54.8%)	595 (58.3%)	57 (33.5%)	< 0.001
6–10	246 (20.7%)	205 (20.1%)	41 (24.1%)	0.230
More than 10	292 (24.5%)	220 (21.6%)	72 (42.4%)	< 0.001
CBD diameter (mm)	11.7 ± 4.2	11.8 ± 4.2	11.0 ± 3.9	0.046
CBD sludge	304 (25.6%)	255 (25.0%)	49 (28.8%)	0.290
CBD stone	694 (58.3%)	606 (59.4%)	88 (51.8%)	0.061
Stone/sludge extraction				
No	325 (27.3%)	265 (26.0%)	60 (35.3%)	0.012
Successful	801 (67.3%)	696 (68.2%)	105 (61.8%)	0.096
Unsuccessful	64 (5.4%)	59 (5.8%)	5 (92.9%)	0.128

(Continues)

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TABLE 1 | (Continued)

Variables	Included patients ($n=1190$)	No PEP ($n=1020$)	PEP ($n=170$)	p
Stone size				
Up to 7 mm	260 (21.9%)	221 (21.7%)	39 (23.0%)	0.155
7–15 mm	273 (22.9%)	236 (23.1%)	37 (21.8%)	0.578
More than 15 mm	161 (13.5%)	149 (14.6%)	12 (7.1%)	0.023
Difficult cannulation	505 (42.4%)	394 (38.6%)	111 (65.3%)	< 0.001
Pancreatic duct cannulation	234 (19.7%)	168 (16.5%)	66 (38.8%)	< 0.001
Sphincterotomy	1075 (90.3%)	912 (89.4%)	163 (95.9%)	0.008
Balloon dilation	1026 (86.2%)	876 (85.9%)	150 (88.2%)	0.410
Biliary stent insertion	363 (30.5%)	304 (29.8%)	59 (34.7%)	0.199

Note: Numerical variables are represented as mean ± standard deviation and categorical variables are represented as numbers (percentage).
Abbreviations: CBD, common bile duct; MDP, major duodenal papilla; NKF, needle-knife fistulotomy; PEP, post-ERCP pancreatitis; TPS, trans-pancreatic sphincterotomy.

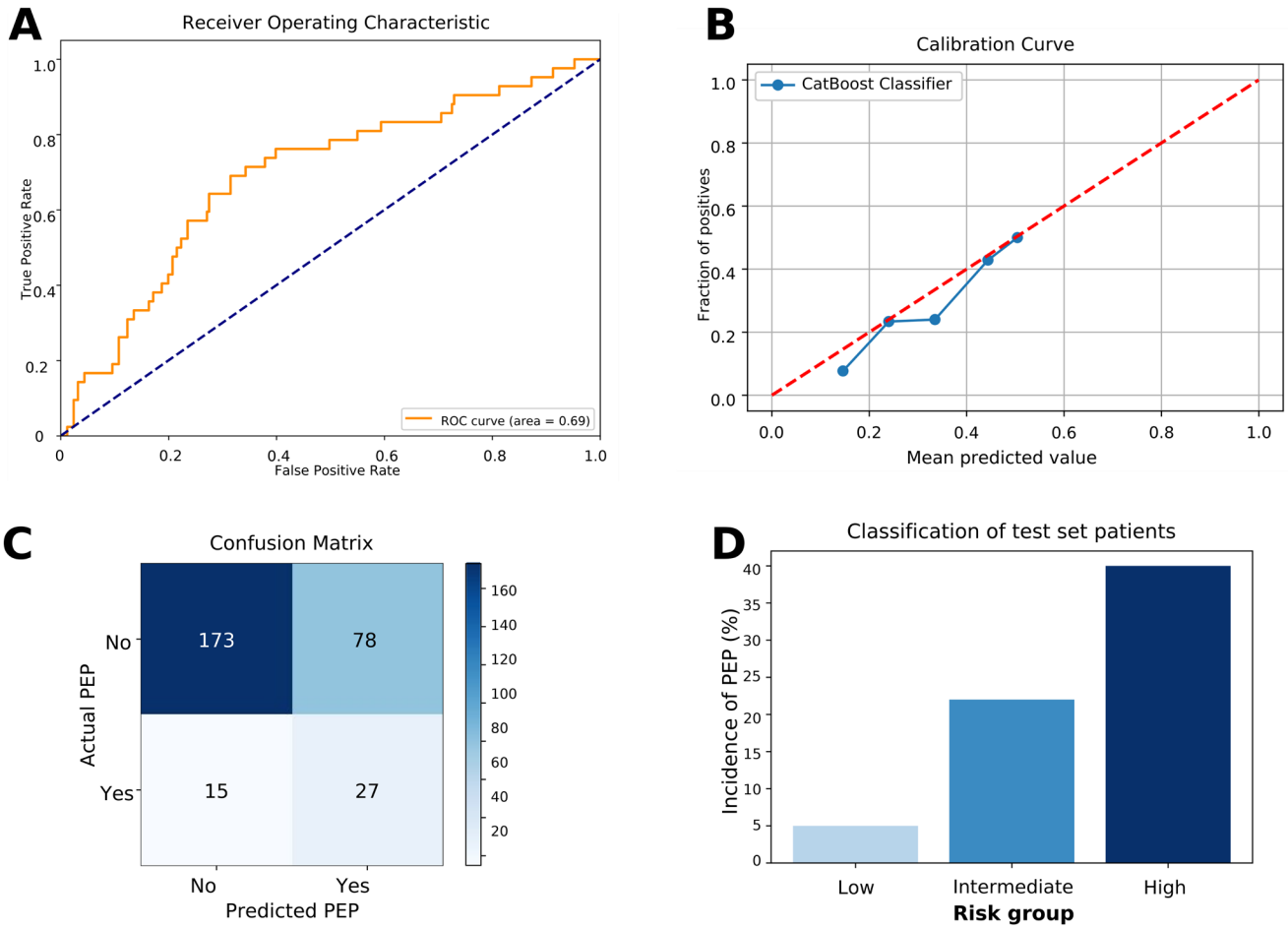


FIGURE 2 | Performance metrics of the best PEP prediction model. Model threshold values were established through analysis of predicted probability score distributions. (A) Area under the receiver operating characteristic (AUC-ROC) curve. (B) Calibration plot comparing predicted versus observed probabilities using the CB model, stratified into four uniform-width probability bins. (C) Confusion matrix illustrating the correlation between model predictions and actual PEP outcomes. (D) Patient stratification across risk categories (low-risk, intermediate-risk, and high-risk), demonstrating increasing PEP incidence with higher risk classification. PEP, post-ERCP pancreatitis; ROC, receiver operating characteristic.

these preventive strategies during the periprocedural phase. Conversely, the ability to identify patients unlikely to develop PEP may allow endoscopists appropriate selection of individuals who can be safely discharged early after the procedure rather than requiring inpatient observation.

In this study, we developed novel ML models to predict PEP. The CB model, utilizing eight features, achieved an AUC of 71.0% in 10-fold cross-validation and 68.8% on the external test set. The most influential predictive features included normal CBD diameter, serum bilirubin levels, younger age, difficult

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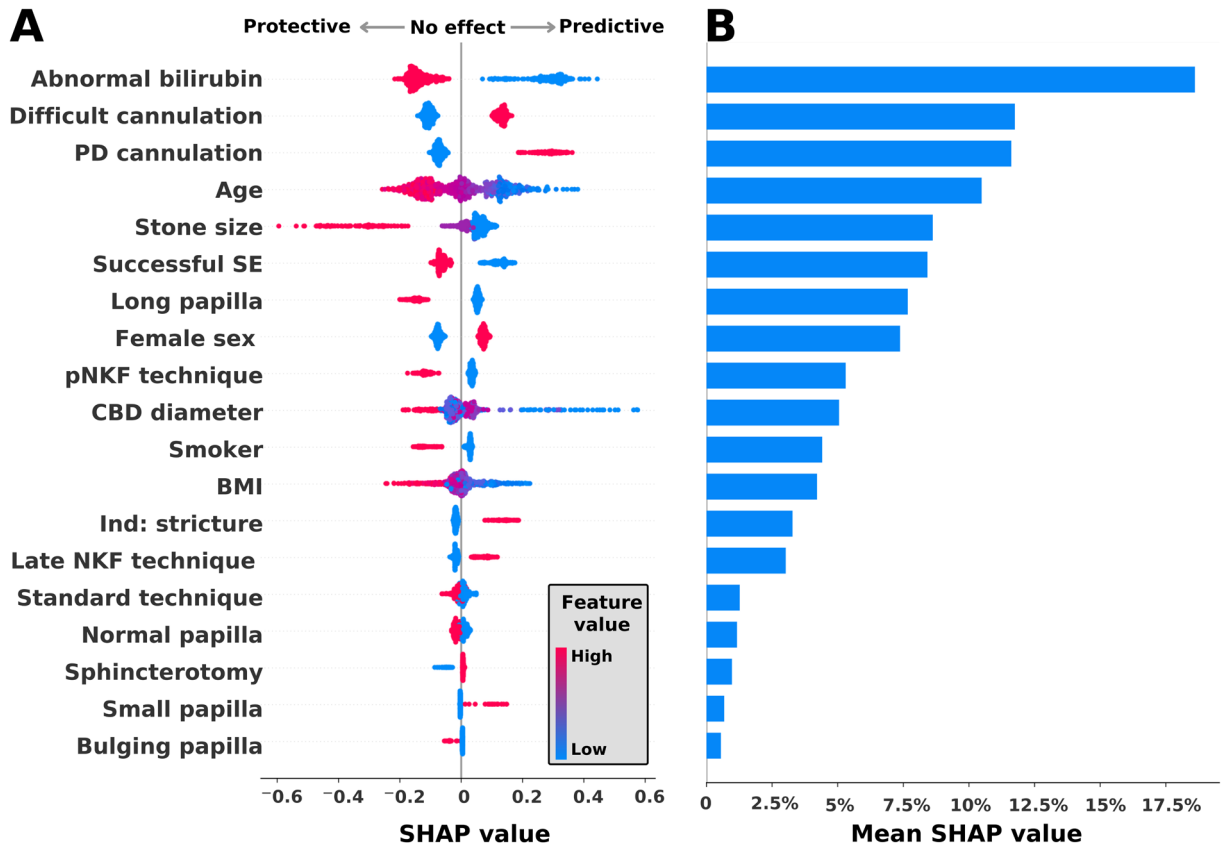


FIGURE 3 | SHAP (SHapley Additive exPlanations) analysis demonstrates clinical features' contribution to PEP risk prediction. (A) Patient-level feature analysis showing individual SHAP values. The horizontal position shows the magnitude of impact on model predictions. The top 10 features and their patterns (asymmetric/symmetric and linear/nonlinear) are: normal bilirubin increases risk (asymmetric); difficult cannulation increases risk (asymmetric); PD cannulation increases risk (asymmetric); younger age increases risk (symmetric and linear); larger stone size decreases risk (asymmetric, nonlinear); successful stone extraction decreases risk (symmetric); long papilla decreases risk (asymmetric); female gender increases risk (symmetric); pNKF use decreases risk (asymmetric); smaller CBD diameter increases risk (asymmetric, linear); and smoking decreases risk (asymmetric). (B) Feature-level analysis presented as a bar plot of mean absolute SHAP values. BMI, body mass index; CBD, common bile duct; Ind, indication; NKF, needle-knife fistulotomy; PD, pancreatic duct; pNKF, primary needle-knife fistulotomy; SE, stone extraction.

cannulation, and PD cannulation. Furthermore, the posterior probabilities generated by the CB model effectively stratified patients into low-, intermediate-, and high-risk groups, with corresponding PEP incidences of 5.7%, 21.0%, and 40.0%, respectively. The model also achieved higher AUC and sensitivity than the experts' opinion model, demonstrating superior performance. It should be noted that the observed incidence of PEP in this study was 14.3%, which is higher than previous reports. This could be related to referral bias, higher rates of difficult and PD cannulation, and including only patients with naïve MDP.

The initial endeavors for predicting and risk-stratifying PEP were performed in 2002 by Friedland et al. using logistic regression models. They developed a scoring system based on four factors including PEP history, pain during the procedure, PD cannulation, and the number of cannulation attempts, and classified patients to low- (<2%), intermediate- (7%), and high- (28%) risk groups [20]. These endeavors have persisted to date. A very recent study from Japan established a pre-procedural scoring system to estimate the PEP risk (SuPER) based on five factors using logistic regression: female sex (+1), naïve papilla (+2), pancreatic calcification on imaging (−2), intraductal papillary mucinous neoplasm (+2), and planned PD procedures (+2). Total scores stratified patients into low (≤0), moderate (1–3), or high (4–7) risk groups, corresponding to PEP rates of 0%–2%, 5%, and 18%–20%, respectively [6]. Nevertheless, the generalizability of all these models has been limited due to biased data, neglecting other patient- and procedure-related factors, and inability to handle interactions between risk factors. For example, applying the SuPER model to the patients of the current study will label all of the patients as moderate risk (score 2 or 3). Such limitations have led to attempts to develop novel prediction models for PEP.

To date, there are several studies that have attempted to predict PEP using ML algorithms. As it could be seen in Table 2, there are significant differences regarding inclusion/exclusion criteria, sample size, training algorithms, and validation methods, resulting in a wide range of performance metrics (AUC from 0.57 to 0.98).

The overall differences in prominent predictive features of PEP among different studies are indicative of a complex mix of methodological, clinical, and technical issues. Firstly, heterogeneity of training data and patient population, as well as their respective baseline features and indications, affects the extent

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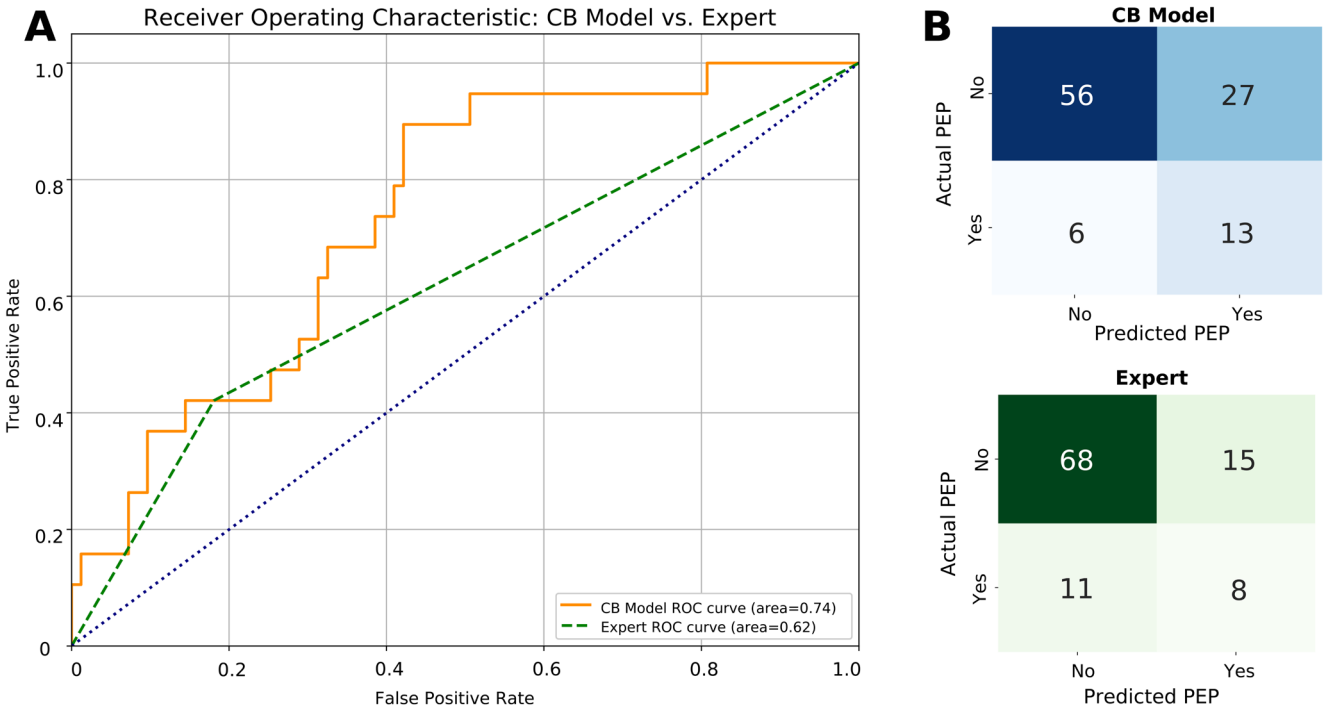


FIGURE 4 | Comparison of predictive performances between the CB model and expert's opinion. (A) Receiver operating characteristic curve. (B) Confusion matrices. CB, CatBoost; PEP, post-ERCP pancreatitis; ROC, receiver operating characteristic.

to which significant each feature will be. For example, lab tests can be strong predictors in a critically ill patient population, but may be less important in an almost healthy group. Second, incoherence in data collection processes directly impacts feature availability and quality. Information related to the procedure, like the number of cannulation attempts, can be precisely documented in prospective single-center registries but possibly missed or documented inconsistently in retrospective multicenter registries, leading to their underrepresentation in model design. Furthermore, this heterogeneity may also result from differences in modeling strategy; feature selection techniques, regularization penalties, and multicollinearity management methods can all influence the importance of each feature. The inclusion of imaging-based features, like papillary radiomics, to certain investigations but not others creates additional variation, especially when certain imaging modalities are not yet routinely available. Finally, the implementation of alternative definitions of PEP, for example, variations in diagnostic criteria for or inclusion thresholds, can subtly modify statistical relationships between features and the target event. Altogether, these factors tend toward the practice of caution in using prediction features across settings, and underscore the importance of external validation as well as harmonization efforts in the development of clinically valid risk models.

We used the SHAP approach to open the black box of ML models and explore how each factor contributed to the model output. Some of these factors have been previously reported as PEP risk factors, including both patient-related (such as female gender and lower age) and procedure-related (such as difficult cannulation and PD cannulation) variables. However, some of these factors have been poorly investigated, with paradoxical results. Among ML models, Chen et al. [24] found no significant association between BMI and PEP. While Brenner et al. [23] reported BMI as the most important variable in their predictive model, they did not report the direction of its effect. In the current study, similar to Archibugi et al. [22], we identified elevated BMI, as well as being a smoker, as protective factors for PEP, which could be due to their fibrotic effect on pancreatic parenchyma [27]. Similarly, while some studies have identified diabetes and alcohol usage as protective factors for PEP [22], others have not attributed significant effects to them [24]. These paradoxical and heterogeneous findings may warrant further investigations through large-scale prospective studies.

Nevertheless, this study has several limitations that must be considered. First, the retrospective design of the study and also performing the procedures by three experienced endoscopists may introduce selection bias and limit the generalizability of the model. Second, and critically, all patients in this study received rectal NSAIDs for prophylaxis. As contraindications to NSAIDs exist in real-world practice, the baseline risk structure for PEP may differ in populations where universal NSAID administration is not feasible, thereby limiting the external validity of our model. Third, patients with prophylactic PD stent insertion were excluded from the study in order to have a more comparable population. Although this group constituted a small proportion of the initial study population (2.5%), they are high-risk patients for developing PEP. Consequently, the performance, calibration, and optimal risk thresholds of our model in this important high-risk subgroup remain unknown and represent a significant limitation to generalizability. Similarly, exclusion of the patients with biliary cannulation failure or altered gastrointestinal anatomy can limit the generalizability of the model. Fourth, although this study developed a predictive model for PEP, it was unable to predict its severity. This was mainly due to limited data, where patients with moderate and severe PEP consisted of only 14.1% of the PEP cases (2% of the included cohort). Fifth, as ERCP

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TABLE 2 | Overview of ML studies for predicting PEP.

First author (year)	Country	Inclusion criteria	Exclusion criteria	Data source	Number of cases for model development	PEP incidence	Learning algorithms	Features used	External validation	Best algorithm and its AUC for test and validation sets	PEP severity prediction	Top predictive features	Conclusion
Takahashi et al. (2023) [21]	Japan	Patients aged ≥ 20 years underwent ERCP for pancreatobiliary disease	Non-naïve papilla, UGI tract reconstruction (except Billroth-I), cannulation failure, ongoing pancreatitis before ERCP	Two hospitals in Japan (one academic, the other regional core hospital)	1159 (615 training, 544 test)	9.9%	RF, LR	Demographic and history, Lab data, Procedural data	Test cohort from a separate center of training cohort	RF, test: 0.77, validation: same as test	No	Albumin, creatinine, biliary tract cancer, pancreatic cancer, CBD stone, total procedure time, PD injection, PGAT, IDUS, bile duct biopsy	The RF model showed better performance than the LR model and was able to significantly classify the patients' risk of PEP
Archibugi et al. (2023) [22]	Spain, Italy, Croatia, Finland, Sweden	Patients aged ≥ 18 years scheduled for ERCP	Not explicitly listed	STARK project (international prospective cohort on using statin and risk of PEP)	1150 (80% training, 20% test)	6.1%	GB, LR	Demographic and history, lab data, procedural data	None	GB, test: 0.67, validation: NA	No	Bilirubin, age, BMI, procedure time, previous sphincterotomy, alcohol use, cannulation attempts, gender, gallstones, use of Ringer's solution, periprocedural NSAIDs	GB model significantly outperformed LR model. The most relevant observed variables were mostly pre-procedural variables, with many of them poorly investigated and worth further studies
Brenner et al. (2024) [23]	International, prospective feasibility study in United States	Same as the original studies	Same as original studies (common: age < 18, pregnancy, recent acute pancreatitis, recent GIB, chronic pancreatitis)	Retrospective analysis of 12 RCTs data	7389 (80% training, 20% test)	8.6%	GBM, RR, RF	Demographic and history, procedural data	Prospective data from 135 ERCPs performed at Johns Hopkins Hospital	GB, test: 0.70, validation: 74	No	BMI, age, GW passage into PD, GW passage into PD > 2, PD injection > 2, cholecystectomy, minor papilla sphincterotomy, difficult cannulation, history of PEP	A novel ML-based PEP risk estimation tool was developed, which can help in prophylaxis selection and its high NPV (95%) can identify patients at low risk who may not require extended post-procedure monitoring

(Continues)

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TABLE 2 | (Continued)

First author (year)	Country	Inclusion criteria	Exclusion criteria	Data source	Number of cases for model development	PEP incidence	Learning algorithms	Features used	External validation	Best algorithm and its AUC for test and validation sets	PEP severity prediction	Top predictive features	Conclusion
Chen et al. (2024) [24]	China	Consecutive patients who underwent ERCP	Age < 18, prior ERCP history, PD stent placement, recent acute pancreatitis, abnormal papilla or confluence features, Failed biliary cannulation, UGI tract reconstruction	Two hospitals in China (between 2019 and 2022)	486 (70% training, 30% test)[a]	33.3%[a]	RF, LASSO, XGB	Demographic and history, lab data, procedural data, radiomic features of papilla	Retrospective data of 96 patients from another center in China	LASSO, test: 0.836, validation: 0.857	Yes	Papilla radiomics score, age, IDUS, cannulation attempts, small papilla, WBC	The radiomic signature of the papilla is a crucial independent predictor of PEP. The papilla radiomics-based model performs well for the clinical prediction of PEP
Xu et al. (2024) [25]	China	Age > 18, naïve major papilla	Patients with missing data, active pancreatitis, failed procedure	Three hospitals in China (one for model development, two for model validation)	1706 (1194 training, 512 test)	6.9%	RF, GNB, KNN, LR, DT, SVM, GBDT	Demographic and history, lab data, procedural data, papilla images	Retrospective data of 210 patients from two other hospitals	RF, test: 0.96, validation: 0.98	No	Papilla image, difficult cannulation, use of NSAIDs, previous pancreatitis, SOD, NKS, therapeutic ERCP, periprocedural fluid solutions, EST, procedure time > 20 min, experience of operator	The study successfully developed prediction models with multistep validation, incorporating various categories of electronic health record data and image-based factors, to assess the risk of developing PEP in ERCP patients

(Continues)

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TABLE 2 | (Continued)

First author (year)	Country	Inclusion criteria	Exclusion criteria	Data source	Number of cases for model development	PEP incidence	Learning algorithms	Features used	External validation	Best algorithm and its AUC for test and validation sets	PEP severity prediction	Top predictive features	Conclusion
Wang et al. (2025) [26]	China	Patients who underwent ERCP for CBD stones between 2015 and 2023	Patients with concurrent PD stones, acute cholangitis, unsuccessful stone removal, residual stones, unsuccessful procedures, or incomplete data	13 hospitals in China (one for model development, 12 for model validation)	2247 (80% training, 20% test)	3.4%	XGB, RF, SVM, NB, GLM, DT	Demographic and history, lab data, procedural data	Separate analysis of the data of 12 centers	RF, test: 0.947, validation: ranging from 0.576 to 0.913 across centers	No	Difficult cannulation, history of pancreatitis, smaller CBD diameter, female gender	The study developed a web application for prediction of PEP in patients with CBD stone who achieved complete stone clearance via ERCP

Abbreviations: AUC, area under receiver operating characteristic; BMI, body mass index; CBD, common bile duct; DT, decision tree; ERCP, endoscopic retrograde cholangiopancreatography; EST, endoscopic sphincterotomy; GB, gradient boosting; GBDT, gradient boosting decision tree; GIB, gastrointestinal bleeding; GLM, generalized linear models; GNB, gaussian naive bayes; GW, guidewire; IDUS, intraductal ultrasound; KNN, *k*-nearest neighbor; LASSO, least absolute shrinkage and selection operator; LR, logistic regression; ML, machine learning; NA, not applicable; NB, naive bayes; NKS, needle-knife sphincterotomy; NPV, negative predictive value; NSAID, nonsteroidal anti-inflammatory drug; PD, pancreatic duct; PEP, post-ERCP pancreatitis; PGAT, pancreatic guidewire-assisted technique without a pancreatic stent; RCT, randomized controlled trial; RF, random forest; RR, ridge regression; SOD, sphincter of Oddi dysfunction; SVM, support vector machine; UGI, upper gastrointestinal; WBC, white blood cell; XGB, eXtreme Gradient Boosting.

[a]The initial population consisted of 2038 patients with a 7.9% PEP incidence; after 1:2 propensity score matching, 486 patients were included for model development.

techniques evolve and become more complex, the model and included features will need to be updated. Finally, and perhaps most importantly, is the low PPV of the model. Although the model was able to identify two-thirds of patients who developed PEP, almost two-thirds of patients predicted to develop PEP actually did not, potentially leading to longer hospitalizations and increased healthcare costs. While previous studies suffer from the same limitation, the clinical and economic implications of such false positives remain a critical concern. To overcome current limitations and advance the clinical applicability of ML in PEP prediction, future research needs to place high priority on the development of more generalizable models. This can be done through prospective, multicenter, large-scale data collection that involves diverse patient populations, different levels of endoscopist expertise, varied PEP prophylactic interventions, and inclusion of high-risk PEP subgroups. Incorporation of deep learning methodologies, particularly those integrating images of the MDP or fluoroscopic data, is a promising avenue for improving predictive discrimination and deserves further investigation. Apart from model construction, clinical utility mandates effortless integration of prediction tools within the ERCP workflow to facilitate real-time risk estimation.

5 | Conclusions

ML is highly promising for prediction of PEP, but current models encounter limitations in generalizability, performance, and clinical implementation. Future studies should focus on multicenter data, inclusion of multimodal data, severity risk-stratification, and real-time application; finally improving patient outcomes and resource utilization in ERCP practice.

Author Contributions

E.A.: study concept and design, acquisition of data, analysis and interpretation of data, drafting of the manuscript, critical revision of the manuscript for important intellectual content. A.S.: study concept and design, acquisition of data, critical revision of the manuscript for important intellectual content. R.R.: acquisition of data, drafting of the manuscript. P.M.: acquisition of data, drafting of the manuscript. S.A.A.S.-N.: study concept and design, analysis and interpretation of data, critical revision of the manuscript for important intellectual content. P.K.M.: study concept and design, analysis and interpretation of data, critical revision of the manuscript for important intellectual content. M.R.Z.: study concept and design, critical revision of the manuscript for important intellectual content.

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We used Meta Llama 3.3 70B Instruct Turbo to enhance the English language of the manuscript and correct grammar mistakes. After rounds of revisions and edits, we provided the draft to the chatbot paragraph by paragraph, using a standardized prompt: "I am a scientific researcher conducting an original study on the prediction of post-ERCP pancreatitis using machine learning. I will give you my manuscript text part by part. Please, improve the English language and correct grammar mistakes. Do not add or delete concepts or information. Just improve

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the English language. Mention all the needed amendments at the end of each paragraph with the reason." One author (E.A.) reviewed and approved the amendments for relevance. All authors reviewed and approved the final version of the manuscript and are responsible for its validity.

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Ethics Statement

This study was approved by the ethics committee of Shahid Beheshti University of Medical Sciences with the ethical code of IR.SBMU.RETECH.REC.1403.411.

Consent

Informed consent was waived by the ethics committee due to the retrospective design of the study.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Forest plot showing odds ratios, confidence intervals, and *p* values for (a) selected and (b) unselected variables by the univariate analysis. **Figure S2:** Heatmap visualizing the Pearson correlation coefficients between selected variables by the previous univariate analysis. **Figure S3:** The contributions of eight features for XGB model 10. **Figure S4:** The contributions of eight features for CB model 16. **Table S1:** Different combinations of features for XGB and relevant metrics (in percent with 95% CIs). **Table S2:** Different combinations of features for CB and relevant metrics (in percent with 95% CIs).

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